Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/18/2006
PI Name:	Wu, J.H. David Ph.D.		
Project Title:	Circadian Rhythm and Control of Hematopoiesis		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14627-0166	**Congressional District:**	28
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2006 per P. Denkins/JSC (7/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Performance Goal No.:			
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During space flights, astronauts are exposed to abnormal light-dark cycles or reduced light, which lead to circadian dysfunction. In long-duration missions, astronauts further suffer from immunohematological disorders and bone loss. In our preliminary studies, we have demonstrated for the first time that a functional circadian clock exists in bone marrow. This discovery has motivated us to seek to establish the link between the circadian clock and hematopoiesis, and hence the link between circadian dysfunction and immunohematological disorders as well as bone loss during space travel.

The overall goal of the project is to: 1) elucidate the molecular mechanisms underlying the circadian rhythms in hematopoiesis; 2) investigate its link with the master clock in the central nervous system; and 3) develop an ex vivo human bone marrow model for testing the potential countermeasure for entraining the clock system and for remedying

Task Description:	circadian dysfunction in bone marrow. Our discovery that a circadian clock exists in bone marrow suggests that the immunohematological disorders encountered in long-duration space flights may be at least partially due to the circadian dysfunction. It is crucial to determine the link between the immunohematological disorders and circadian dysfunction before effective countermeasures can be devised. The study will lead to a better understanding of the link between circadian dysfunction and immunohematological disorders as well as bone loss, an issue of vital importance for long-duration space travel. The results of this research may also benefit those who suffer from circadian dysfunction due to their shift-work schedule on ground or travel across time zones.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research may benefit those who suffer from circadian dysfunction due to their shift-work schedule or travel across time zones on ground.
Task Progress:	The circadian oscillator is composed of interacting feedback-loops formed by both positive- and negative-regulators. In mouse, the negative-regulators include three Period genes (mPer1, mPer2, and mPer3) and two Cryptochrome genes (mCry1 and mCry2). Expression of the negative-regulators is driven by the positive-regulators, such as CLOCK and BMAL1, which form heterodimers via their PAS domains and trans-activate their target genes. Our recent discovery of the bone marrow clock provides a plausible molecular basis of circadian rhythms in hematopoiesis. To further elucidate the function of circadian clock in bone marrow, we examined the effect of circadian gene expression in a murine bone marrow progenitor cell line and a tetracycline-inducible expression system. Tetracycline induction of gene expression in the stably transfected cell line was confirmed by Real-time PCR and western blot. Furthermore, induced expression of the clock positive elements promotes differentiation of the progenitor cell line. While the progenitor cells were going through differentiation, the cell cycle activities were inhibited. These results indicate that circadian clock modulates differentiation of hematopoietic cells and that circadian dysfunctions experienced by astronauts during space flights may have adverse effects on the hematopoietic and immune system. Considering that bone cells including osteoblasts and osteoclasts are generated in bone marrow, the bone-forming process may also be affected. Further studies will elucidate the potential roles of the circadian clock in immunohematological disorders and bone loss experienced in space travel and provide rationales for countermeasure development.
Bibliography Type:	Description: (Last Updated: 01/30/2019)
Abstracts for Journals and Proceedings	Wu JHD, Chen CY. "Circadian modulation in hematopoiesis." NASA Cell Science Conference, Galveston, TX, February 2005. Abstracts, NASA Cell Science Conference, Galveston, TX, February 2005. p. 39. , Feb-2005
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Articles in Peer-reviewed Journals	Panoskaltsis N, Mantalaris A, Wu JH. "Engineering a mimicry of bone marrow tissue ex vivo." J Biosci Bioeng. 2005 Jul;100(1):28-35. Review. https://doi.org/10.1263/jbb.100.28 ; PubMed PMID: 16233847 , Jul-2005